Residual FlashSAC Grounding of 4D Human-Object Motion onto G1+Dex3

V2D Challenge 2026 - Track 2 Technical Report

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1. Introduction

Track 2 of the NVIDIA Isaac Video-to-Data (V2D) Challenge asks whether a fixed humanoid embodiment can replay 4D human-object interaction (HOI) as a learned policy. Embodiment and simulator are given: Unitree G1 with Dex3 hands in FlashCHORD (Newton / Warp / JAX). The scored output is the object pose the policy actually produces, not a reconstructed or copied reference. All three input tiers are measured against the same clean multi-view object trajectory (30 episodes, 22,989 frames at 30 Hz, 29 unique objects).

We entered Tier 1 (clean SOMA-X / multi-view captions) and Tier 2 (the official synthetically corrupted reference). We did not enter Tier 3. Trajectories are policy rollouts from evaluate_policy.py with world_count=1 and VOC scale 0. We did not use Tier 1 captions on Tier 2, and we did not submit reconstructions as rollouts.

2. Approach

Pipeline. Released LeRobot episodes are converted to SOMA-X folders, retargeted to G1+Dex3 with the official soma_to_g1 path, and trained with FlashSAC on the ReconBody objective. Support geometry (USD cylinders/cubes) is enabled only when the retargeted USDA actually contains it; empty support files are left off. One policy is trained per episode. Evaluation writes an object-pose parquet per episode; the official packer converts those poses into the released world frame with one rigid transform and emits a single CSV for that tier's four Kaggle boards.

Learning. The legal FlashSAC recipe uses 2048 parallel worlds. We ran three budgets on the same retargeted motion: a short 0.5M-step probe, a 16.8M-step compressed-VOC schedule, and the official 350M-step VOC curriculum (recon_body_flash_sac). Residual FlashSAC acts on SONIC joints with scale 0.15. A later hard-episode curriculum (recon_body_flash_sac_object) raises object-position and palm-orientation weights for clips that still drop or twist. An object-orientation geodesic term was added to ReconBody for a final near-miss pass; those policies were not mixed into the counted Kaggle file.

Selection. For each of the 30 episodes we keep the rollout with the best CHORD gates (SP-SR then MP-SR) and lowest MPPE among the short, 16.8M, 350M, and hard runs. The counted Tier 1 file is this mix (22×350M, 4×16.8M, 2×short, 2×hard). Tier 2 uses one 16.8M FlashSAC policy per episode trained on the noisy reference only.

3. Experiments and Results

Training ran on RTX PRO 6000 GPUs, about two hours per 350M episode. Eval is deterministic (actor mean). Local packer scores match Kaggle. Counted submissions:

Entry	AUC up	SP-SR up	MP-SR up	MPPE cm down
Tier 1 350M uniform	0.409	0.467	0.267	17.98
Tier 1 mix (final)	0.434	0.533	0.300	13.04
Tier 2 16.8M (final)	0.257	0.167	0.033	49.15

On the counted Tier 1 mix, 16/30 episodes pass SP-SR (≤10 cm and ≤0.5 rad) and 9/30 pass MP-SR (≤3 cm and ≤30°). The mix recovers two failures that uniform 350M missed: tall-bar-stool pull (episode 21) reaches SP after the object/palm retrain (5.7 cm, 13.5°), and squeegee (episode 28) drops from 41 cm to 7.3 cm MPPE. Four 16.8M rollouts beat their 350M counterparts (episodes 0, 1, 7, 22), so longer training is not uniformly better.

Residual errors are concentrated. Baseball bat and cyan water bottle remain near 1 m MPPE across every budget — thin, rotating objects that the G1 grasp never stabilizes. Vacuum, easter

basket, and snack box sit at 20–39 cm. Several near-misses are millimetres or tenths of a degree from a gate (trash-can drag 3.12 cm vs the 3 cm MP line; book 28.78° vs the 28.6° SP line). Those clips are orientation- or position-gated, not collapsed.

Tier 2, trained only on the noisy reference, keeps the same embodiment and packer but loses most of the success-rate mass (SP 5/30, MP 1/30). That gap is the Track 2 measurement: the same FlashSAC recipe is much less tolerant of upstream pose noise than of clean SOMA-X. We did not recover a better Tier 2 trajectory from video, and we did not peek at Tier 1 captions.

4. Conclusion

A legal G1 FlashSAC stack, per-episode policies, and a best-of mix across training budgets produce a valid Track 2 policy-rollout submission. Scaling to the official 350M VOC schedule is the main lift; mixing shorter runs and a targeted object/palm retrain recovers additional SP/MP mass and cuts mean MPPE from 18 cm to 13 cm. The remaining failures are a small set of unstable objects, not a global retargeting bug. Code used at evaluation time is nvidia-isaac/video_to_data commit 0593d80, plus the per-episode FlashSAC launch and mix scripts in our training repository.

Industry innovation (judging areas)

Economic viability. One FlashCHORD recipe, no Isaac Sim/Lab in the scored loop, and about two GPU-hours per 350M episode. Best-of mixing reuses finished rollouts instead of retraining the full set.

Data flywheel. Each new budget (0.5M, 16.8M, 350M, hard) is another candidate per episode. The mix table is the flywheel: later runs replace a clip only when they beat the stored gates and MPPE.

Complexity reduction. Fixed G1+Dex3, official retargeter, official evaluator and packer. No custom simulator, no reconstructed meshes in the scored file, one CSV per tier.

Innovation. Per-episode residual FlashSAC with a CHORD-aware mix, plus a stronger object/palm curriculum on the clips that still fail after 350M, rather than a single global policy or a copied reference trajectory.

Kaggle (final): Track 2 Tier 1 57625045 / 57625046 / 57625050 / 57625052 (AUC / SP-SR / MP-SR / MPPE). Track 2 Tier 2 57508092 / 57508094 / 57508095 / 57508096. Runtime commit: https://github.com/nvidia-isaac/video_to_data/commit/0593d80de5d39ab6f56f73ed16ca464ba5053499. FlashCHORD image flash_chord:latest.